

Lecture : Introduction to Data Science Tools
Term : Fall 2024
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Computer? It is an electronic device that is capable of performing 4 main tasks: Input (keyboard, mouse, mic, touchpad, scanner, camera, sensors,), Processing, output (screen, speakers, printer, ...), and storage.

ASCII: American Standard Code for Information Interchange ($127 = 2^7 - 1$ numbers correspond to characters on a keyboard. For example, $A = 65_{\text{Base}=10} = (01000001)_{\text{Base}=2}$, $B = 66$, ..., $a = 97$, $b = 98$, $c = 99$

UNICODE $\rightarrow 2^{16}$

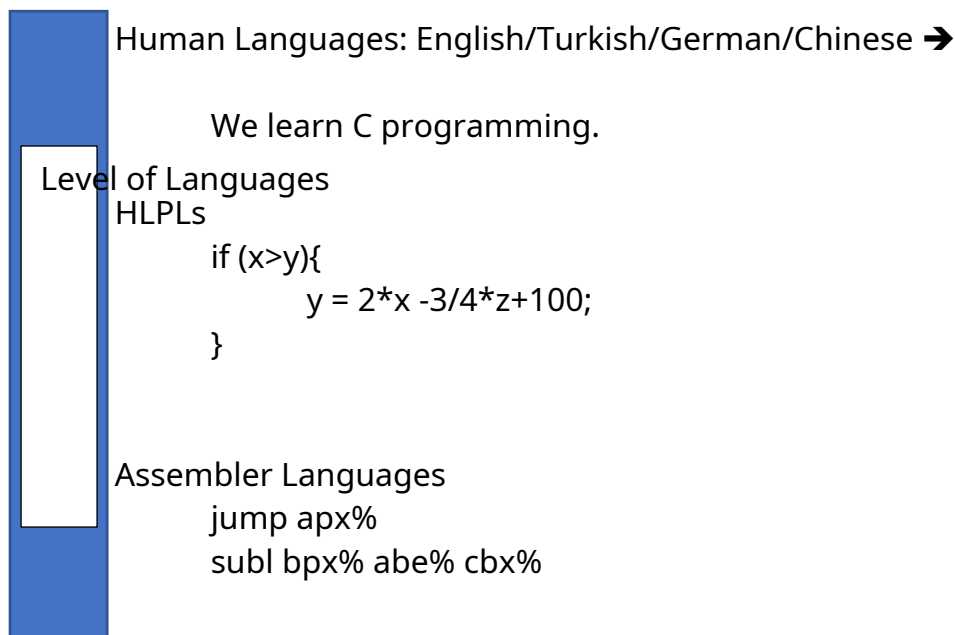
Speed of a computer = 4 GHz $\rightarrow 4 \times 10^9 \rightarrow$ roughly the number of instruction per second is $\rightarrow 1/(4 \times 10^9) = 10^{-9}/4 = 0.25 \times 10^{-9}$ seconds

Program? A set of instructions for computer to perform a specific task. $y = 2 * x - 3/4 * z + 100$; Computers are FAST but STUPID, human being on the other hand is SLOW but SMART (Einstein).

All Programs run on memory (RAM, L2, L3, or L1/primary (cache), and registers)) including the Operating System (OS).

Compiler? Is a program that converts a code written in a High-Level Programming Language (HLPL) to machine language.

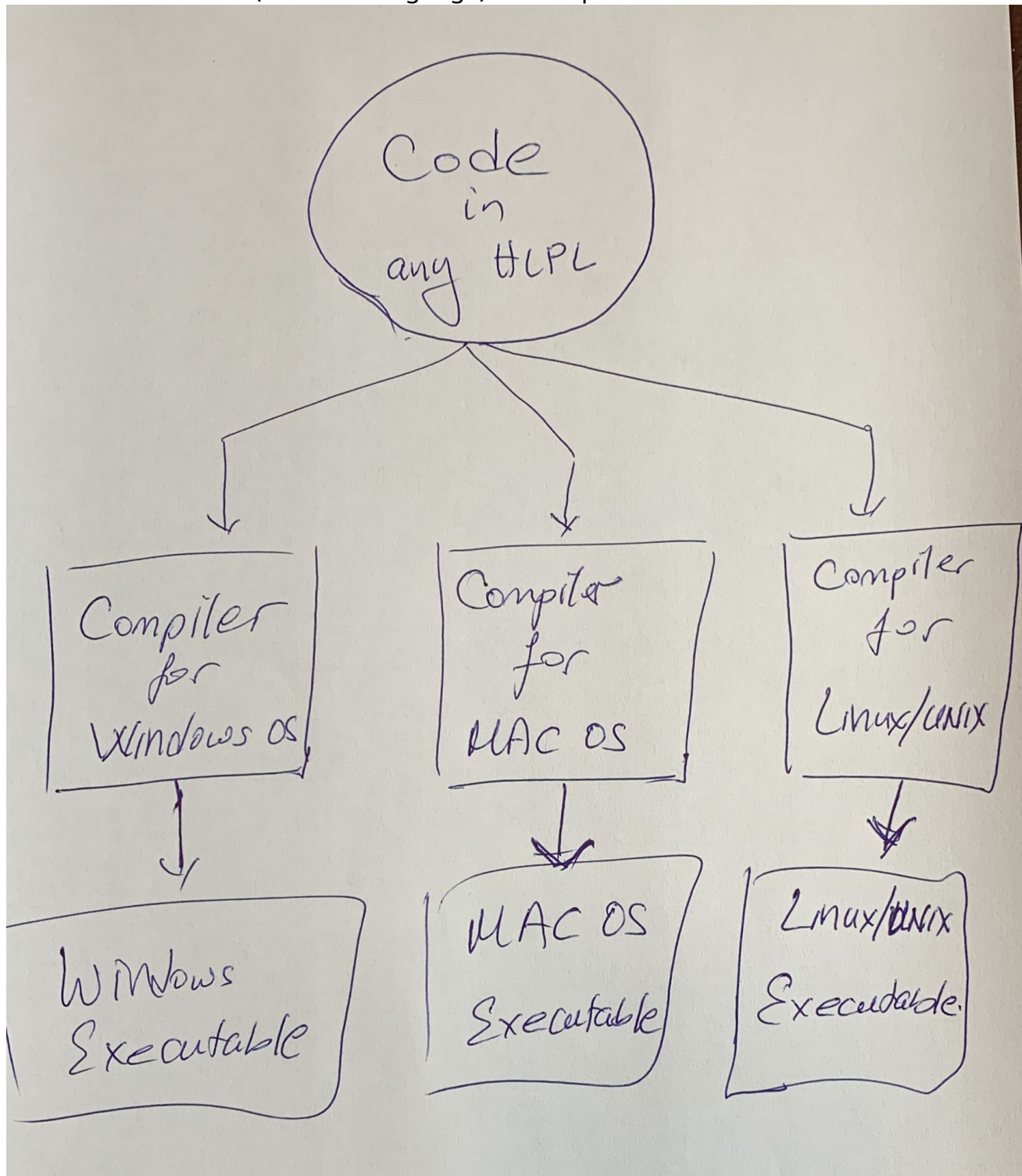
Popular HLPLs: C, C++, Java, Python, JavaScript, R, Ruby on Rail, Delphi, Pascal, Fortran, Ada, Scala, Basic (QBasic),



movl bpx% abe%

Machine Language: 0000101010101010110

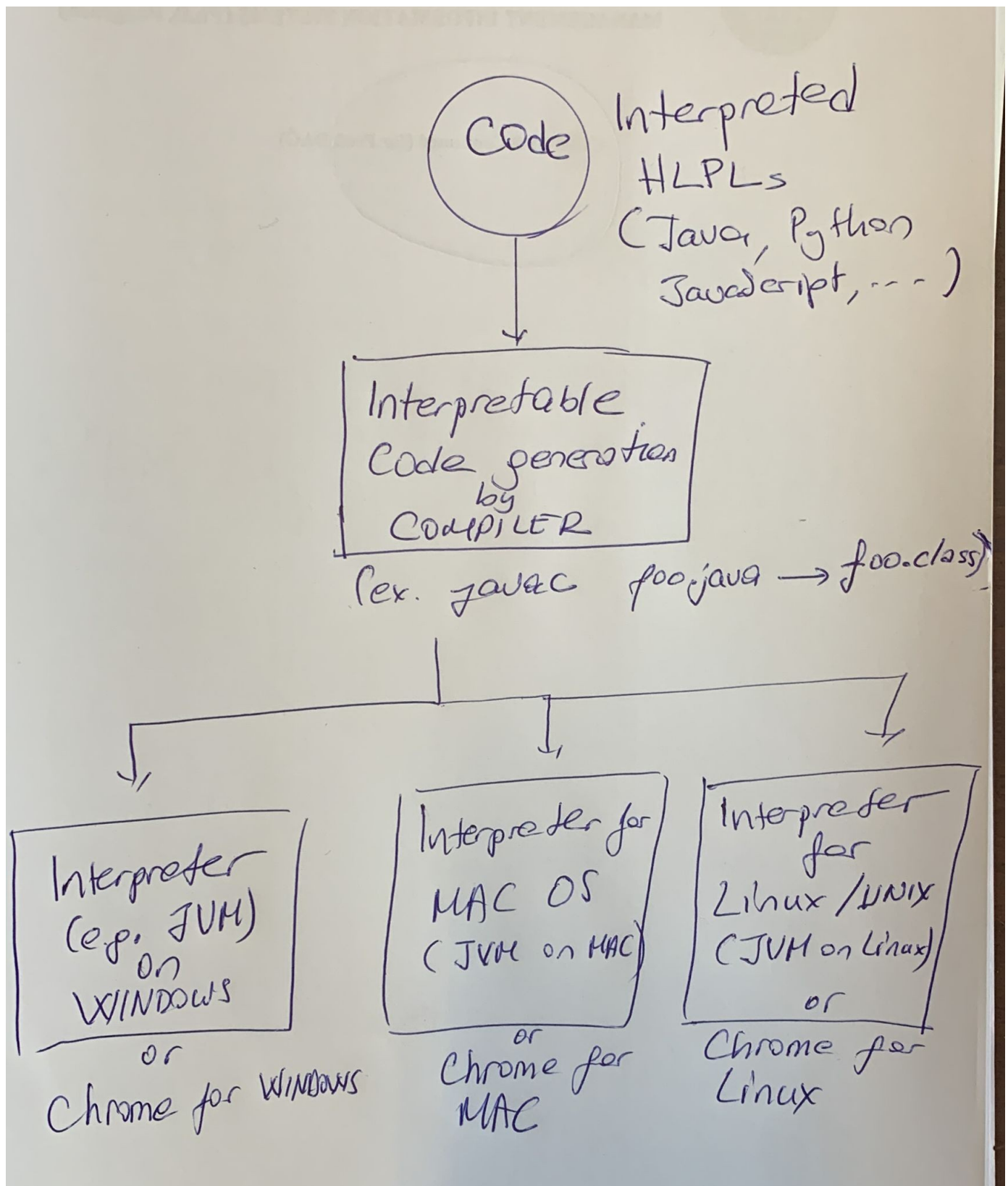
Compiled HLPL (for example C, C++, Fortran, Pascal,, etc.). After compilation a self-runnable executable (machine language) code is produced.



Code (in any HLPL) is written in a text editor (such as Notepad, or vi/vim/pico/nano (on Linux or MAC OS)) or using an IDE (Integrated Development Environment) such as Visual Studio, .Net).

IDE = (Advanced) Text Editor + Compiler + Debugger

Interpreted HLL (for example Java, JavaScript, Php, Perl, Python, etc..). After compilation an intermediate file (for example BYTECODE for Java) is produced and this intermediate file requires an interpreter, such as Java Virtual Machine (JVM) on the related platform to run.



Constructs of a structured HLPL:

- Tokens (!, =, <, >, ==, ==, x, I), expression (5, x>0, 2*x-5, i), variable (also called identifier)

- Statements
- Control structures (Conditional Execution, or Branching in the code)
 - o if
 - o if-else
 - o switch-case-break
- Loops (Repetitions): Doing the same thing repeatedly until the controlling expression returns False

o for

```
// for (expression1; expression2; expression3)
for (initialization; controlling_expression;
    changing_controlling_variable_value){
    statement(s);
}
```

o while

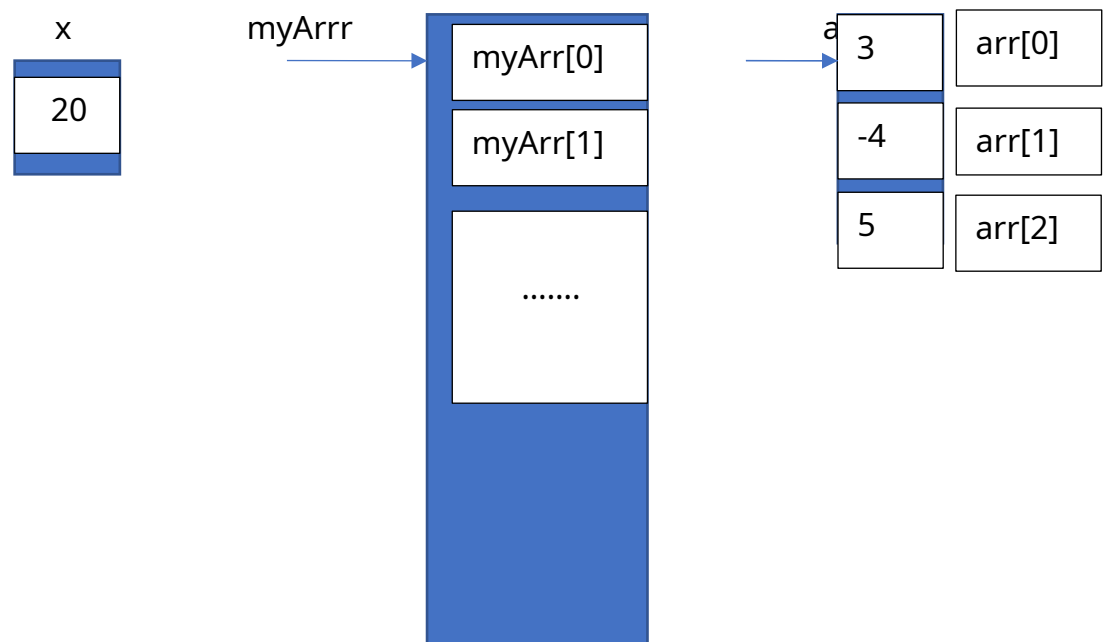
```
while(expression){
    statement(s);
}
```

o do-while

```
do{
    statement(s);
} while(expression);
```

- Arrays → Ability to store many values of the same type with a single variable name

```
int x=20;
int myArr[10], arr[] = {3,-4,5}
```



myArr[9]

- (User Defined) Functions

```
return_type function_name(arguments){  
  
    function_body (statement(s));  
    return returned_value; // or nothing if void  
  
}
```

- Files

Storage mechanisms for storing text, binary material, pictures, videos: write, read and append

++ means increase the value by one

-- means decrease the value by one

➔ ++i ➔ pre increment
i++ ➔ post increment
--i ➔ pre decrement
i-- ➔ post decrement

int x = 10, z = 5

int y = 2*x+/--z; ➔ Y = 2*10/4

printf("x =%d, z=%d, and y = %d\n", x, z, y);

x	z	y
10-11	5 4	5

C divides the operators into the following groups:

- Arithmetic operators (+, -, /, *, %, ++, --)
- Assignment operators %(=)
- Comparison operators (<, <=, ==, >, >=, !=)
- Logical operators (&&, ||, !)
- Bitwise operators (&, |, !)

Precedence of operators: PEMDAS (**P**arenthesis, **E**xponent, **M**ultiplication, **D**ivision, **A**ddition, Subtraction), then left to right